Appendix A - Abrolhos Marine Park (Clio Bank) Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. SPZ = Special Purpose Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
Jun 2025	2025-06 Clio-Bank stereo-BRUVs	BRUV	SPZ	72	60	60	57

Benthic ecosystem extent

Observations

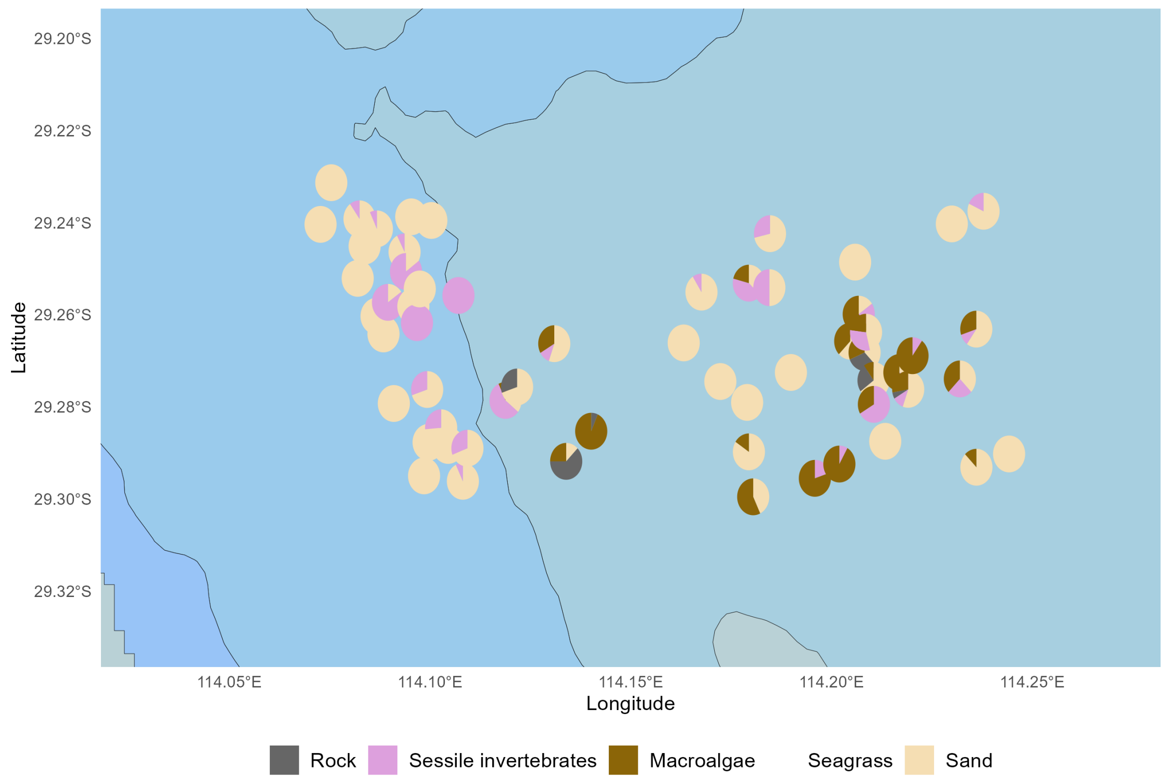

Figure 1: Benthic classes observed in spatially balanced stereo-BRUV imagery, visualised as spatial pie charts. All classes scored are shown, including those too rare to model. Commonwealth marine park boundaries are shown.

Dominant benthic type

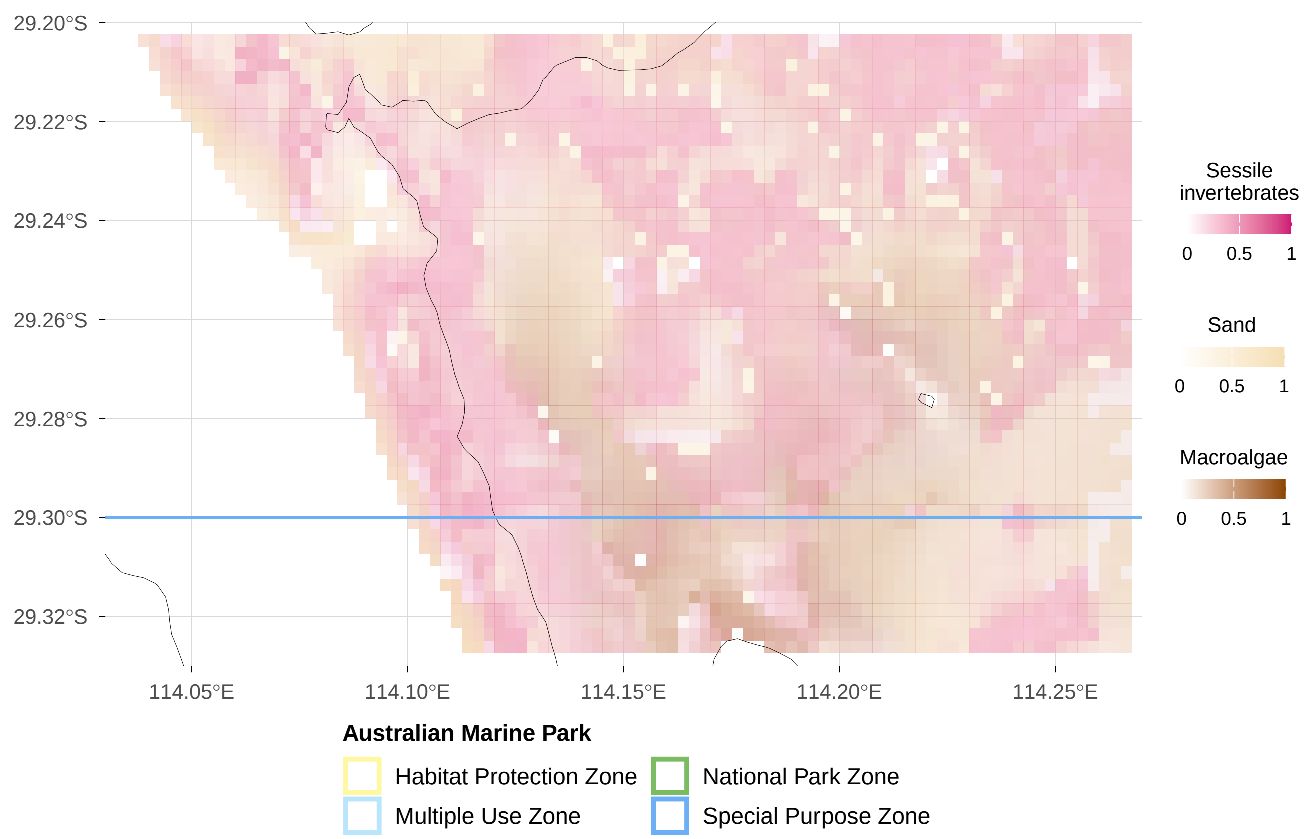

Figure 2: Predicted dominant benthic type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.

Benthic distribution and probability

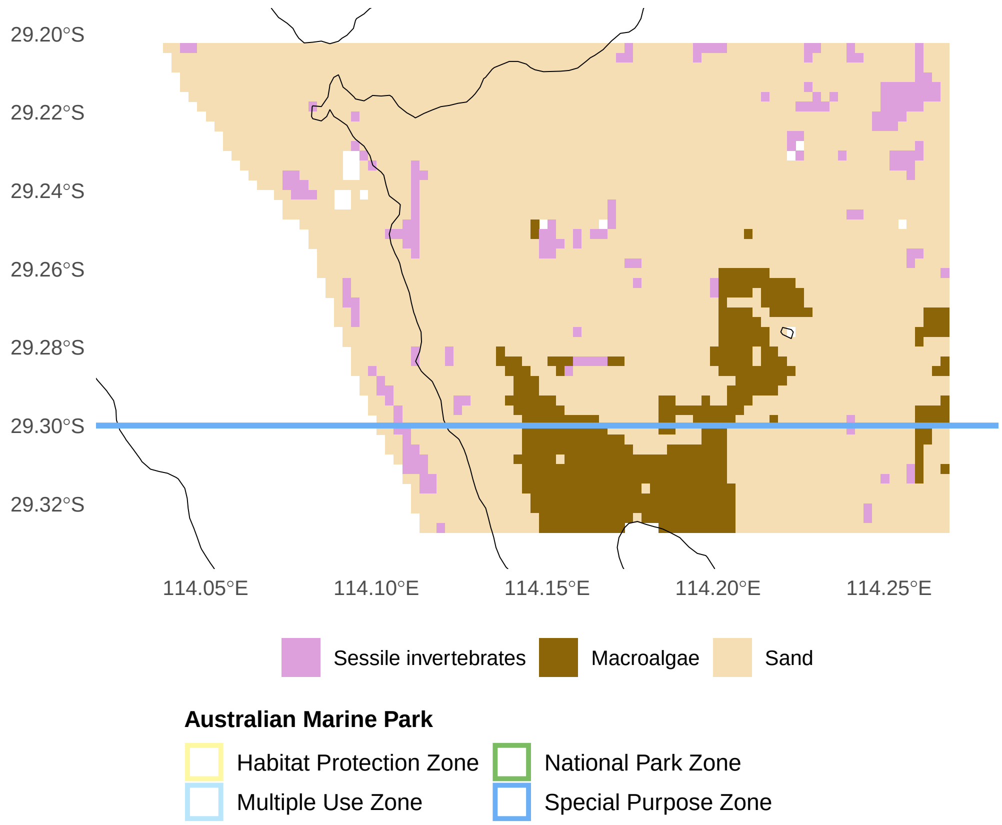

Figure 3: Predicted benthic distribution probability from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.

Functional reef distribution and probability

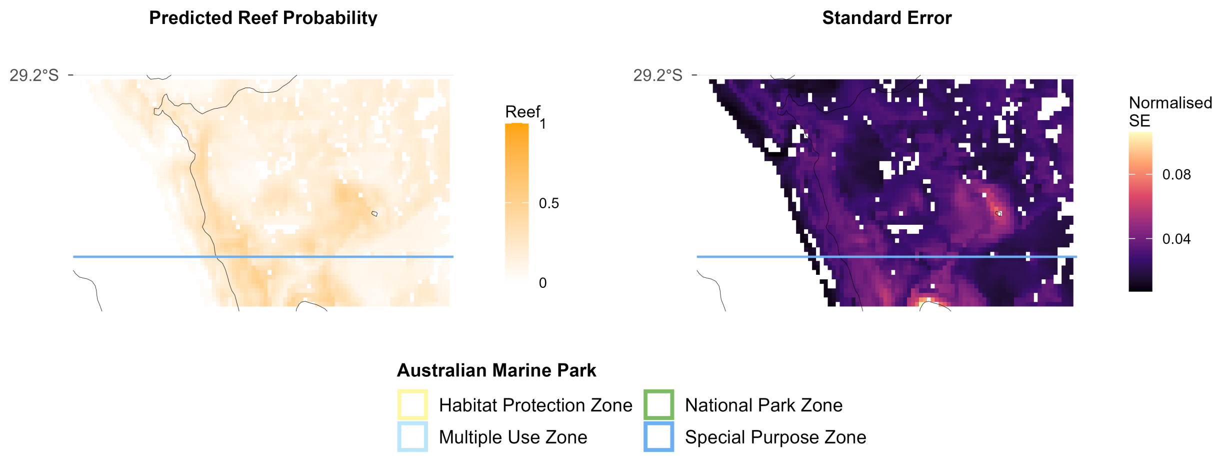

Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.

Benchmarks of benthic natural values

Sand

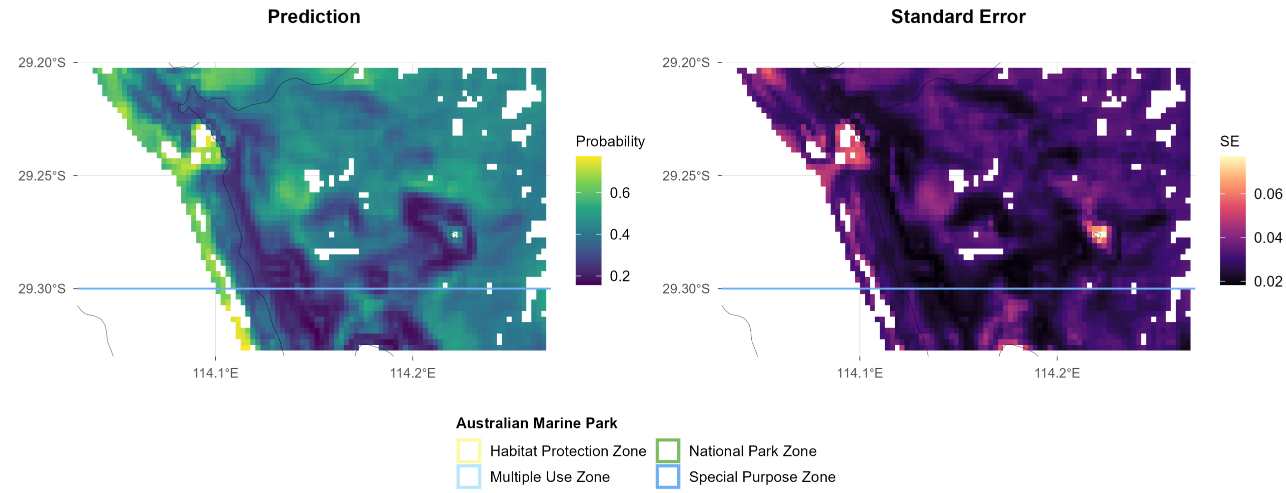

Figure 5: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown.

Macroalgae

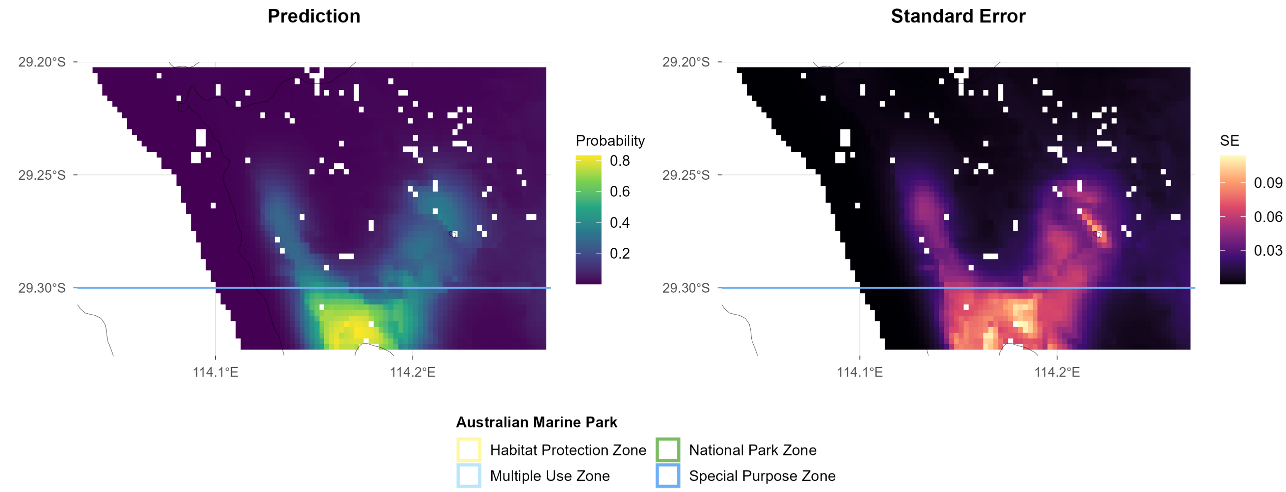

Figure 6: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown.

Sessile invertebrates

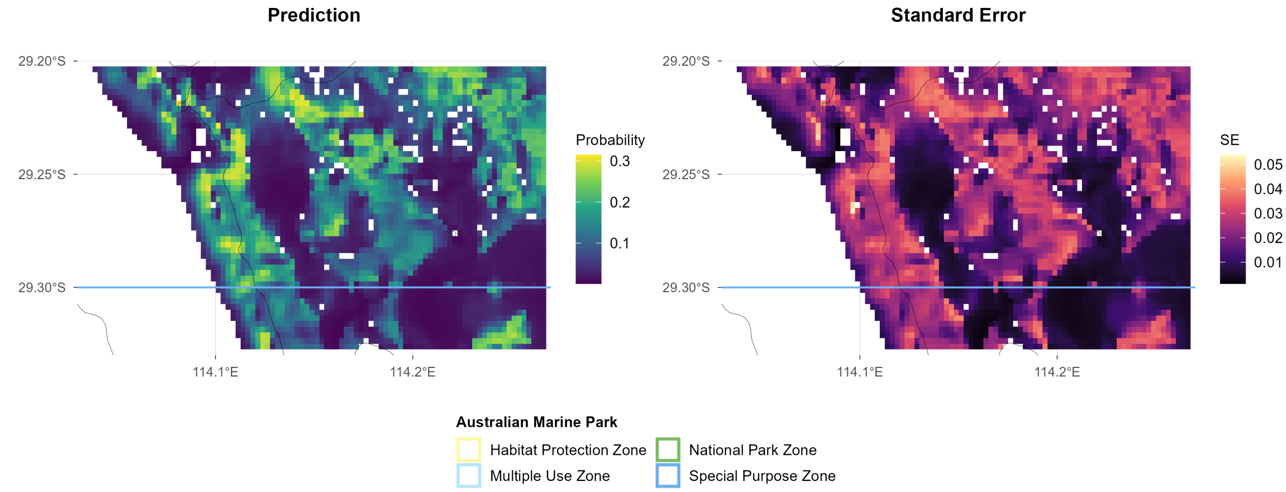

Figure 7: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown.

Benchmarks of demersal fish natural values

Survey effort

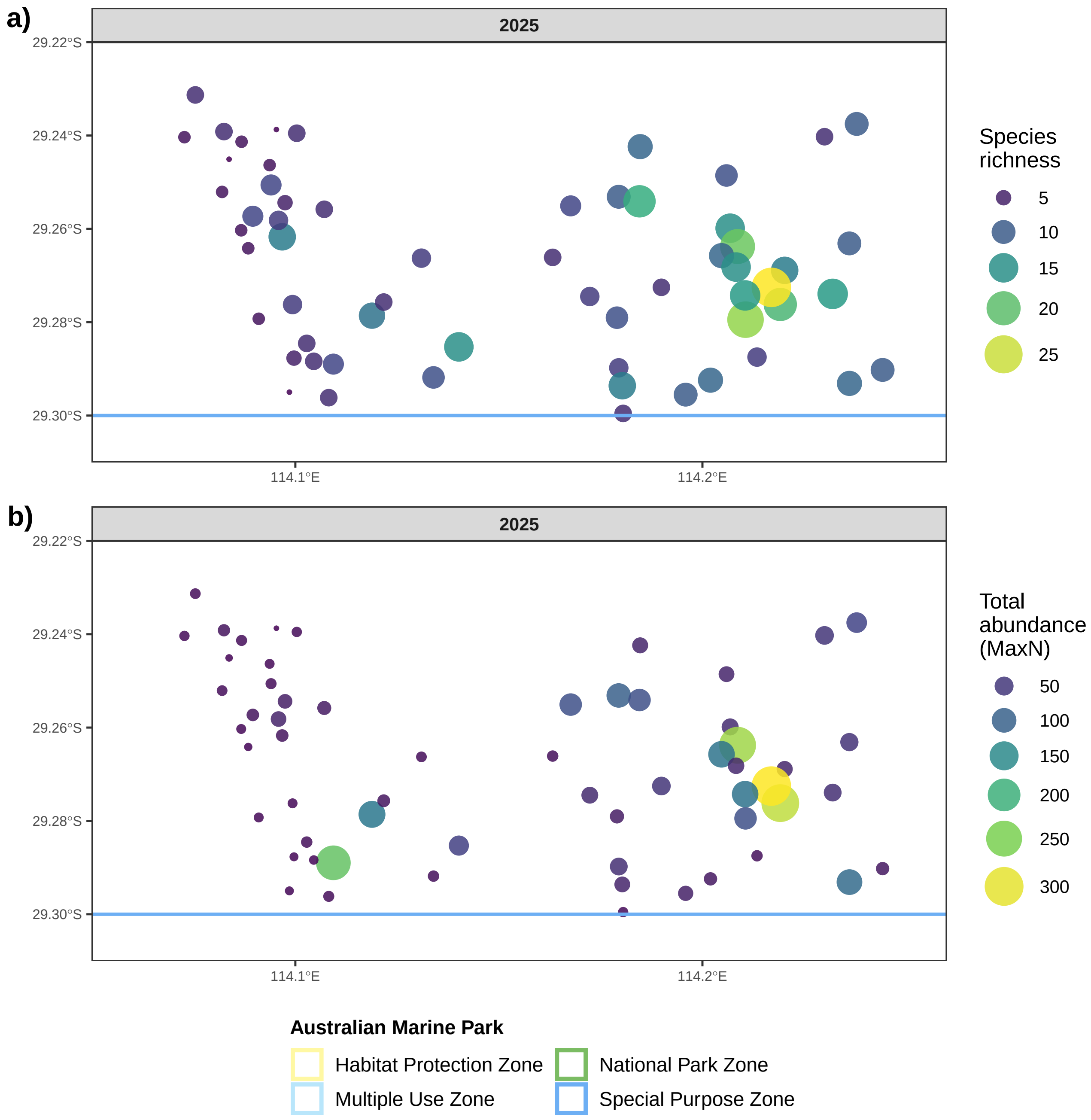

Figure 8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop. Commonwealth marine park boundaries are shown.

Species richness

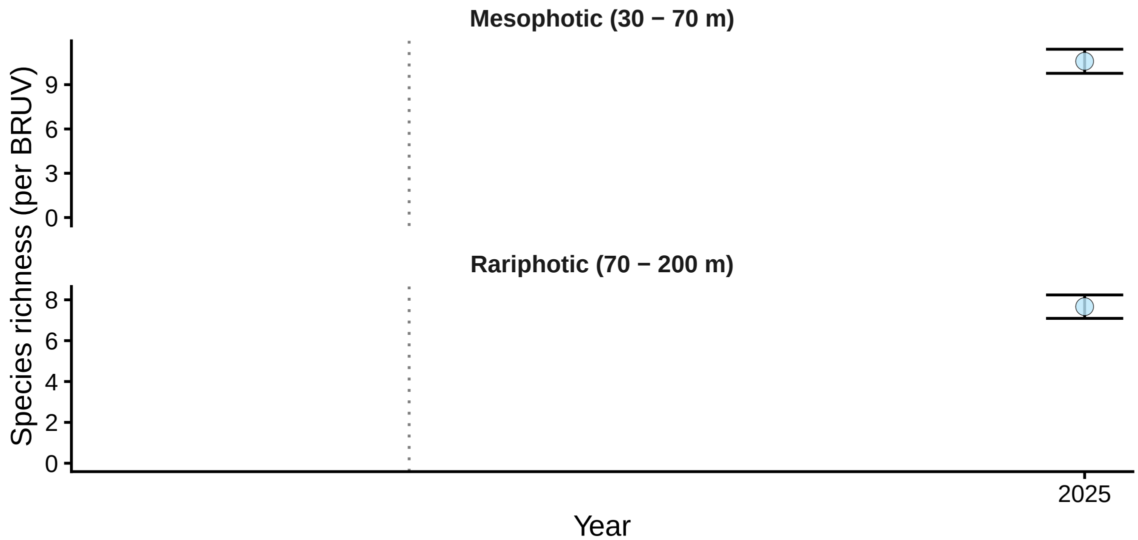

Figure 9.1: Demersal fish - Species richness: benchmark plot by depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). All deployments fall within the Special Purpose Zone, so no zone contrast is shown.

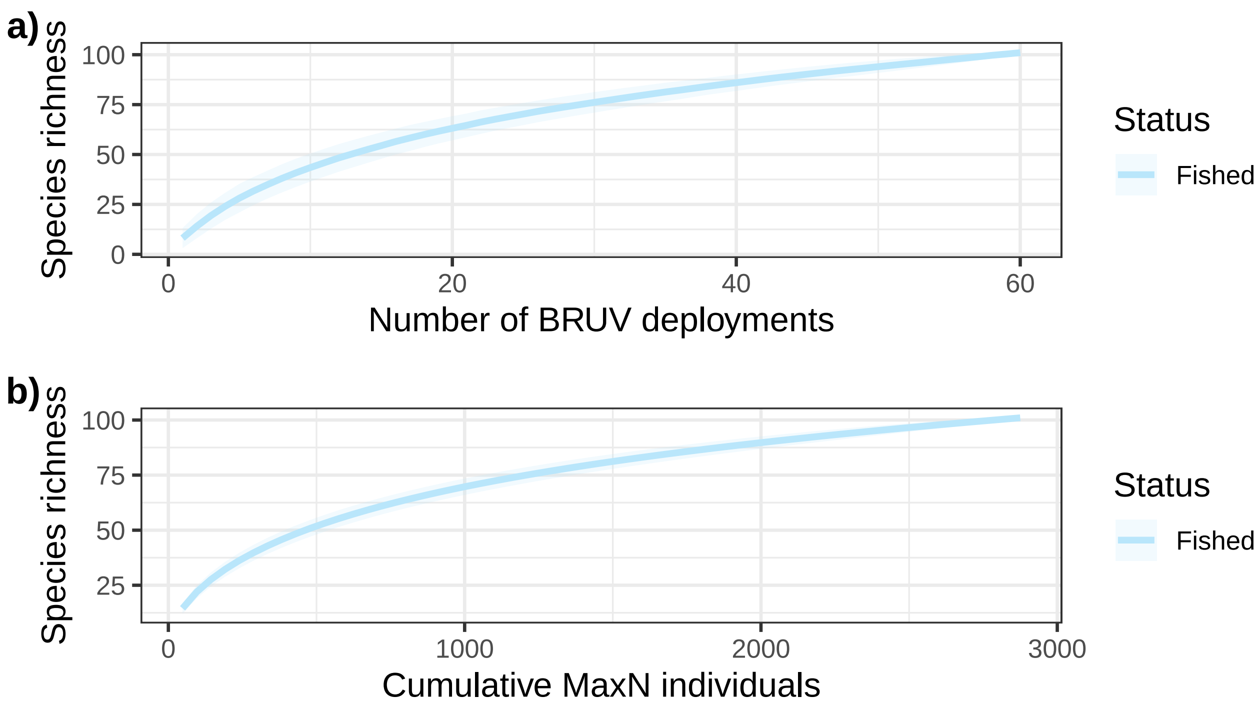

Figure 9.2: Demersal fish - Species richness: species accumulation curves from stereo-BRUV groundtruthing. a) sample-based detection/non-detection, b) individual-based rarefaction. ABMP = Abrolhos Marine Park (Commonwealth).

Species richness

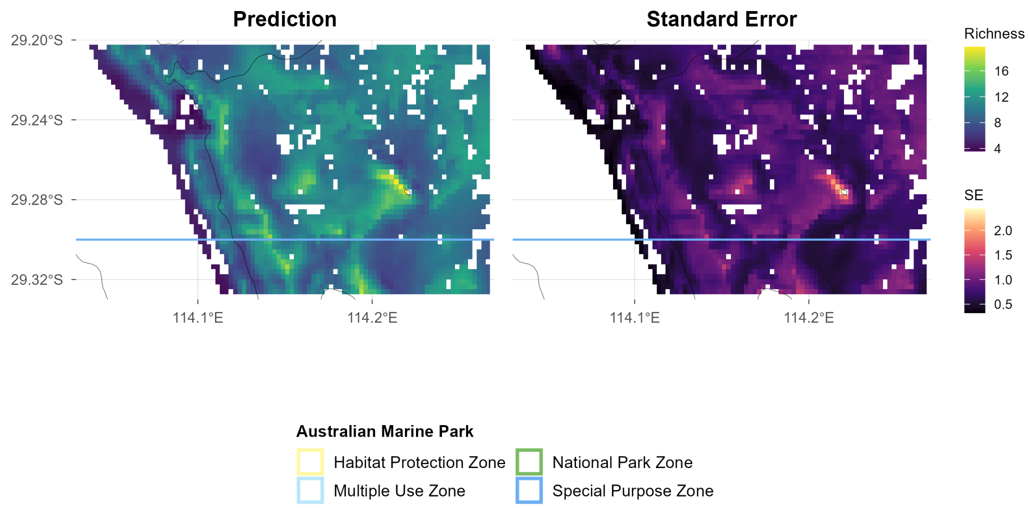

Figure 9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.

Total abundance

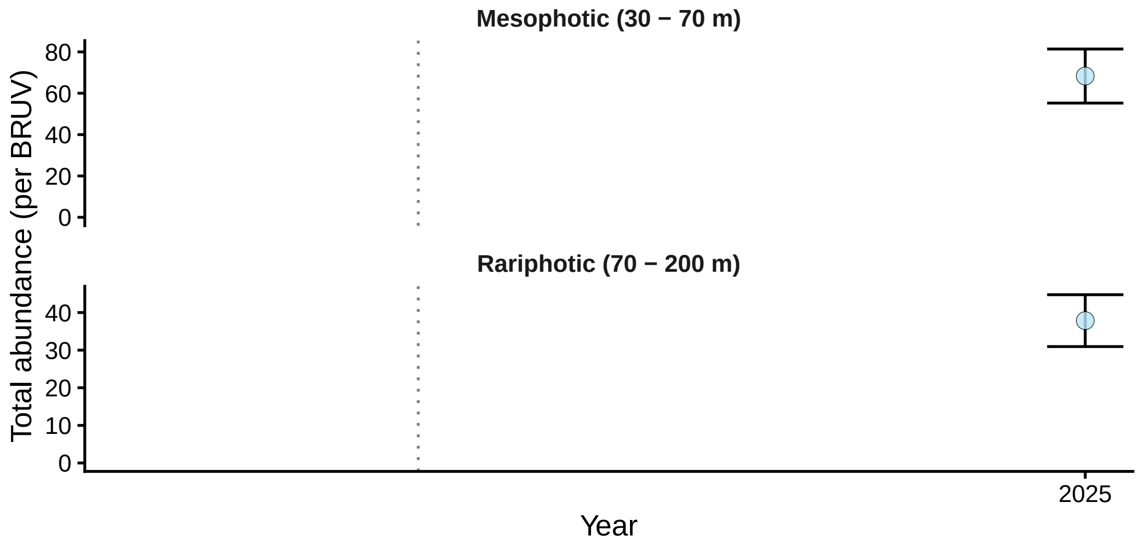

Figure 10.1: Demersal fish - Total abundance: benchmark plot by depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). All deployments fall within the Special Purpose Zone, so no zone contrast is shown.

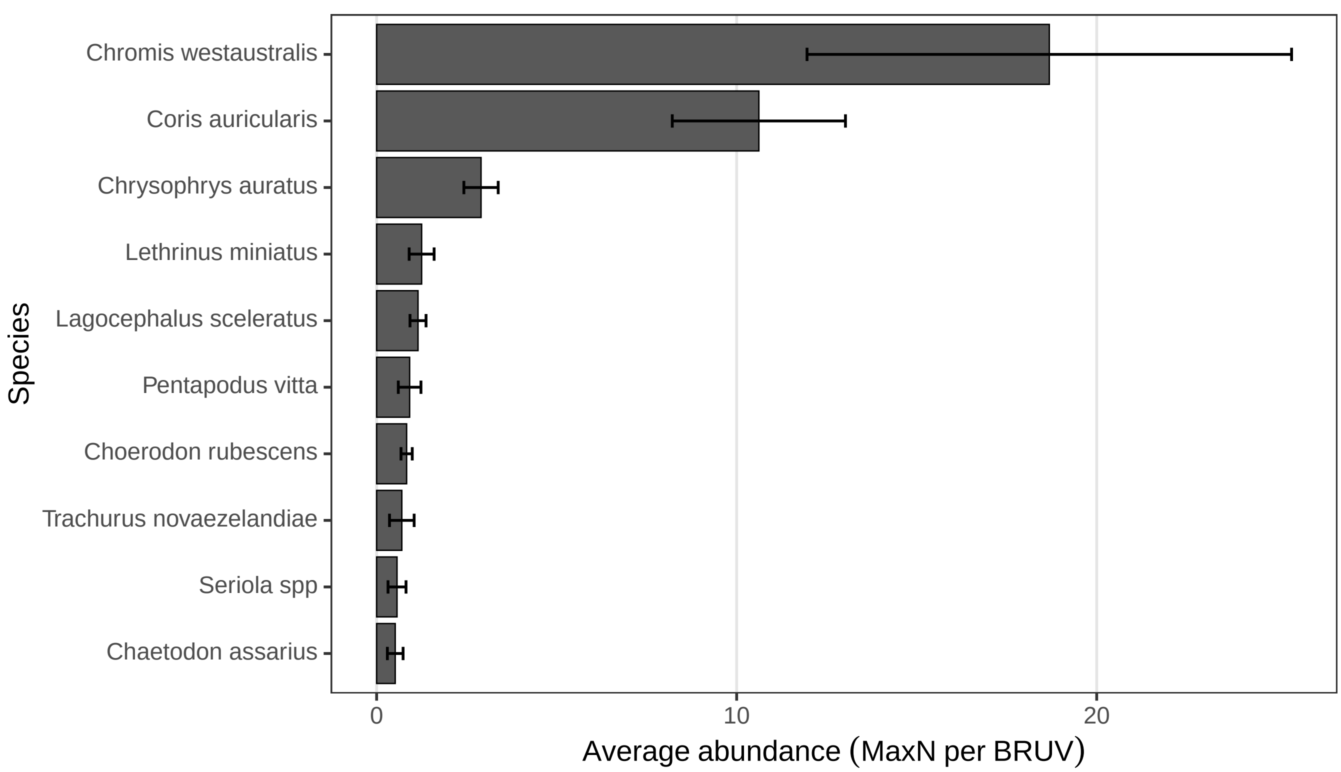

Figure 10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Total abundance

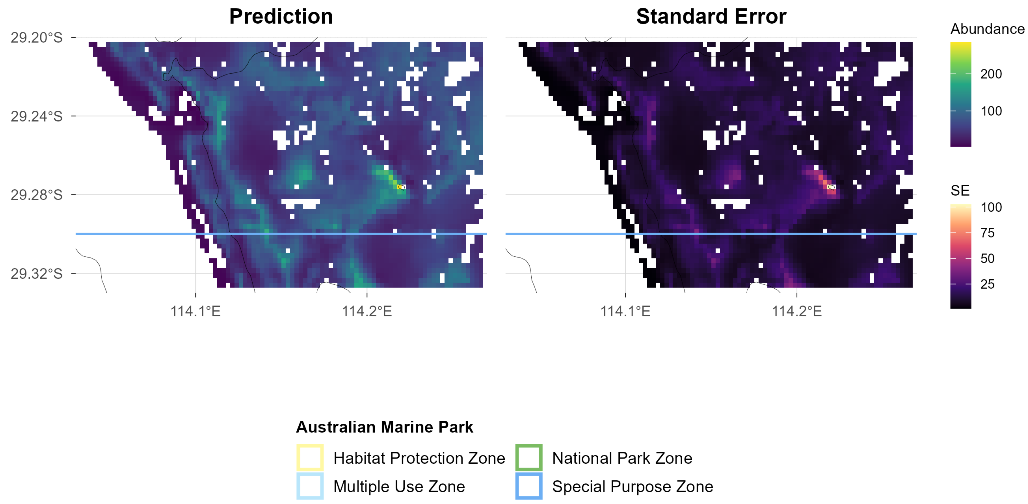

Figure 10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.

Large Reef Fish Index*

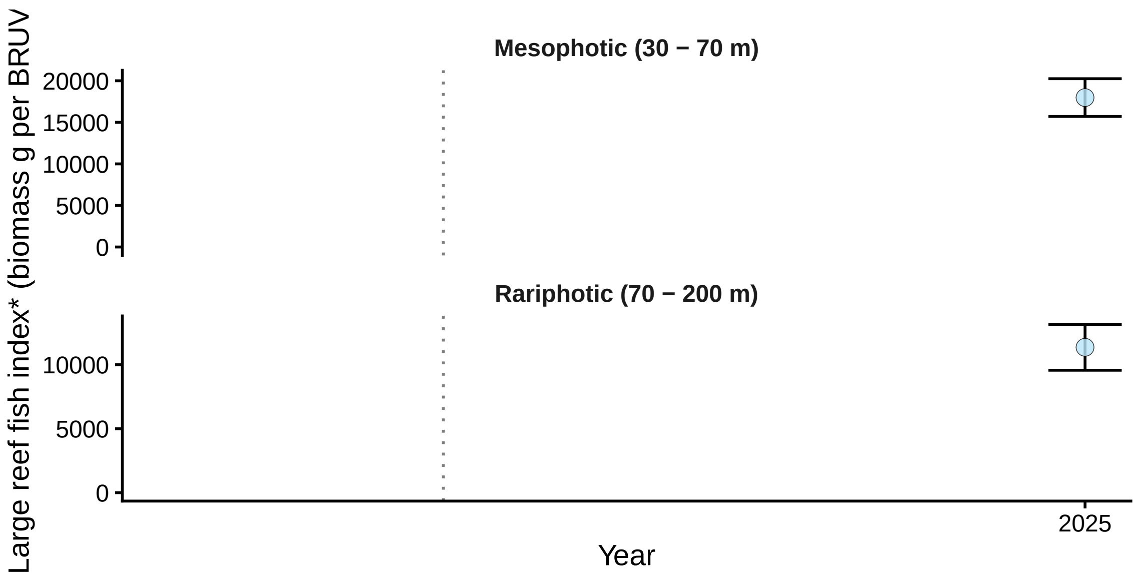

Figure 11.1: Demersal fish - Large Reef Fish Index*: benchmark plot by depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). All deployments fall within the Special Purpose Zone, so no zone contrast is shown. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

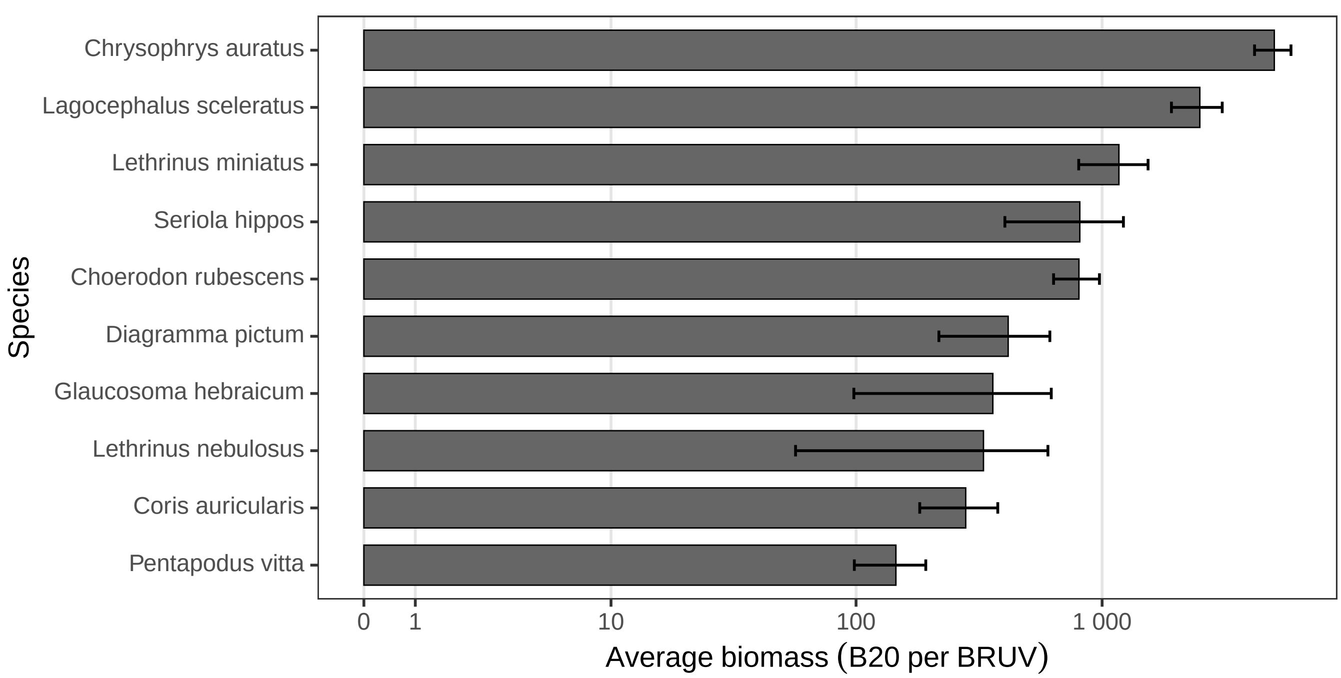

Figure 11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Large Reef Fish Index*

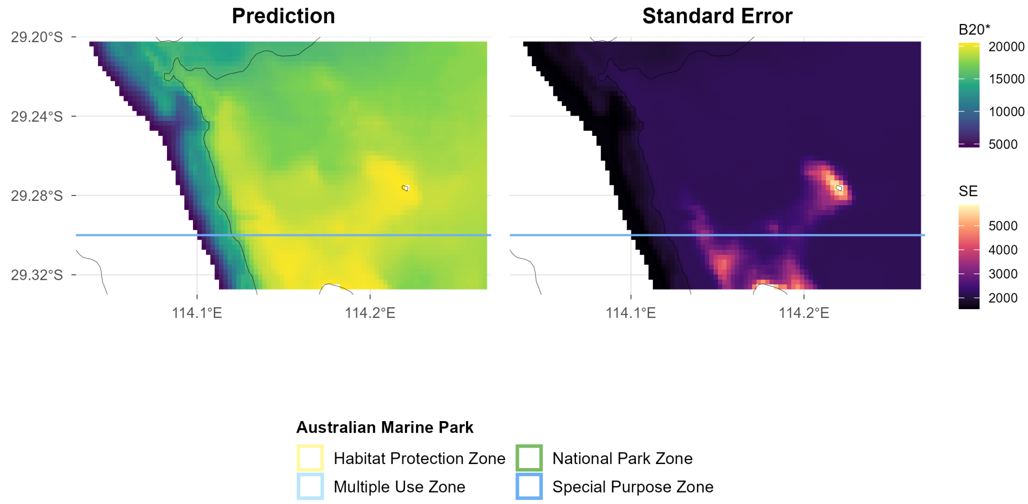

Figure 11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.

Reef fish thermal index

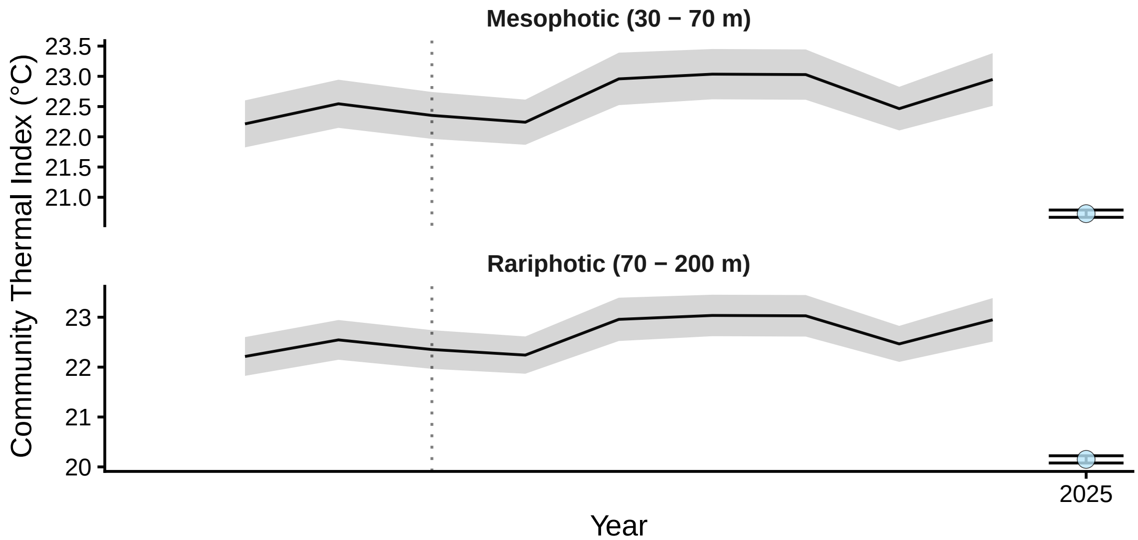

Figure 12.1: Demersal fish - Reef Fish Thermal Index: benchmark plot by depth contour, with annual sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). All deployments fall within the Special Purpose Zone, so no zone contrast is shown.

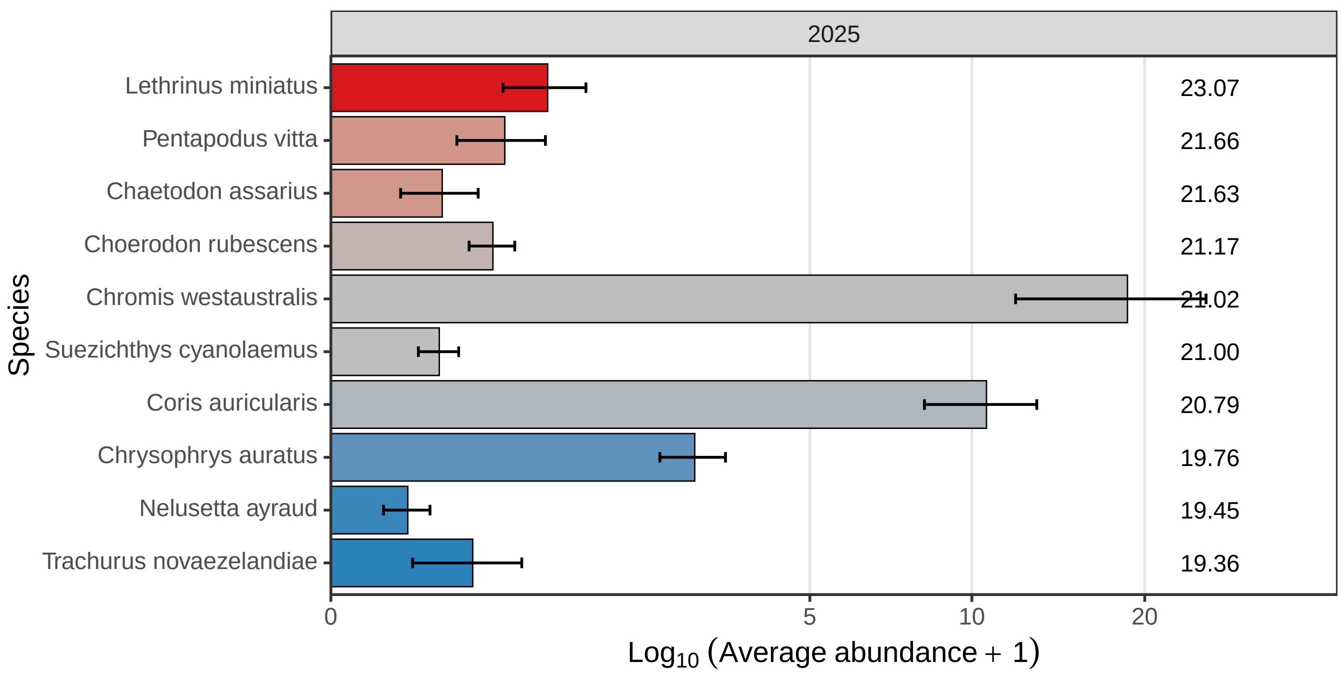

Figure 12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of the top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Reef fish thermal index

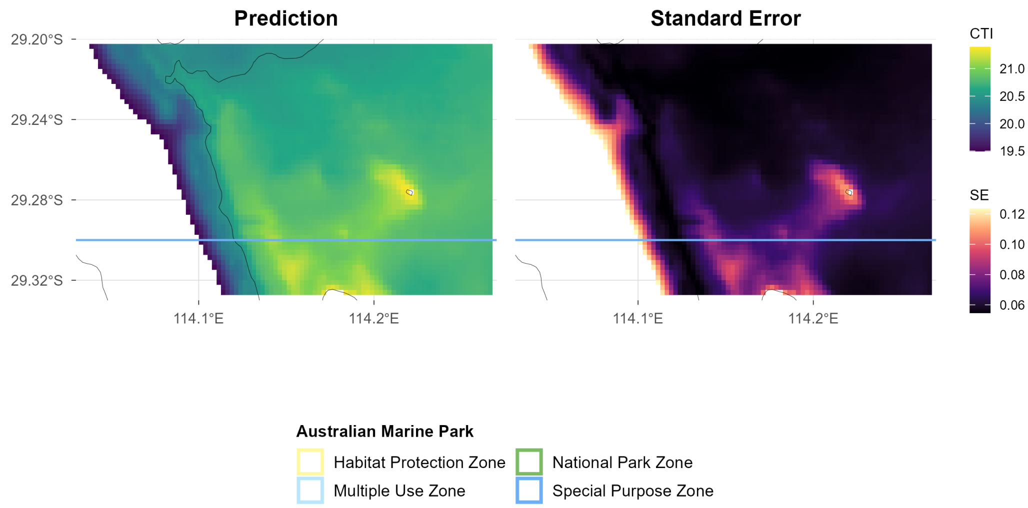

Figure 12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown.